Propaganda Detection Using Fine-Tuned DeBERTa-v3-large with RoRA Adapters and Active Learning

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1. Abstract

This project constructs and validates a social media–based dataset for fine-grained propaganda detection. While existing datasets are primarily derived from news articles, real-world propaganda is increasingly prevalent in informal platforms such as Reddit. Using a DeBERTa-v3-large classifier with RoRA adapters, 1,500 out of the 13,118 Reddit posts were filtered through a dual-LLM consensus pipeline to produce 173 high-confidence annotations across 14 techniques. When combined with the SemEval-2020 dataset, the inclusion of Reddit samples improves Macro F1 to 0.817.

2. Objective

Labeled datasets for fine-grained propaganda detection remain scarce, particularly for social media where language is informal and politically charged.

This project aims to:

- Collect and curate a Reddit-domain propaganda dataset across 14 technique categories
- Build a quality-controlled labeling pipeline using LLM ensemble consensus
- Validate dataset quality through a fine-tuned DeBERTa classifier on SemEval-2020

3. Methodology

Base Dataset: SemEval-2020 Task 11 — 4,904 sentence-level annotations across 14 propaganda techniques used as the seed training set.

Reddit Collection: 13,118 Reddit posts were scored by model uncertainty. The 1,500 most uncertain samples per round were selected as candidates.

LLM Ensemble Labeling: Each candidate was independently labeled by Llama 3.1:8b and Gemma 2:9b. Only samples where both agreed were accepted. Round 1: 173 accepted (11.5%). Round 2: 6 accepted (0.4%), confirming saturation.

Label Format: Each sample is a 14-dimensional multi-hot vector with soft coverage scores where techniques overlap.

Model Architecture

The backbone is DeBERTa-v3-large (442M parameters), chosen for its disentangled attention mechanism which separates content and position encoding — making it effective at detecting context-dependent manipulation techniques like loaded language and appeal to fear.

RoRA adapters are injected into the last 12 of 24 transformer layers, targeting the query, key, value, and dense projection matrices (72 modules total, r=32, α=64). Only 8.14M of 442M parameters are trained, keeping the backbone frozen and preventing catastrophic forgetting. RoRA's key distinction from standard LoRA is its $\alpha/\sqrt{r}$ scaling, which produces more stable gradients.

The final [CLS] token is passed through a 14-output linear classifier with BCEWithLogitsLoss for multi-label prediction. Each label uses an individually calibrated decision threshold tuned for 90% precision.

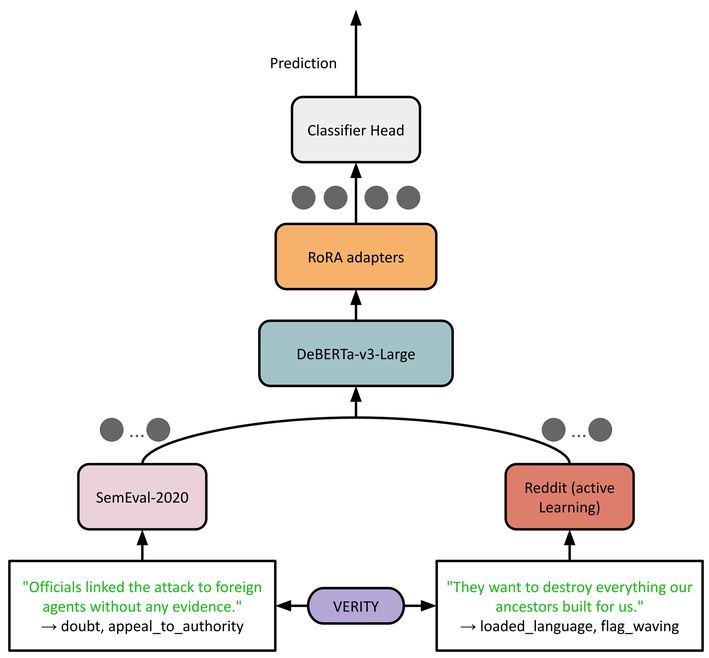

Figure 1. *VERITY propaganda detection pipeline. Author's own work.*

4. Results

Phase	Description	Macro F1
Phase 1	Full fine-tune — SemEval + Reddit	0.541
Phase 2	RoRA fine-tune — SemEval only, backbone frozen	0.766
Phase 3 (AL Round 1)	+173 LLM-labeled Reddit samples	**0.817**
Phase 4 (AL Round 2)	+6 new samples (stopping criteria met)	0.790

Table 1. *Model performance across training phases. Results from author's own experiments.*

Dataset quality was validated by retraining the classifier on the combined SemEval and Reddit dataset after each collection round. Progressive improvement in Macro F1 confirms that the collected Reddit samples carry genuine propaganda signal consistent with the SemEval annotation standard. The 0.4% acceptance rate in Round 2 confirms dataset saturation, marking the natural endpoint for data collection.

Technique	Gold Spans	Span F1
Doubt	493	10.19%
Loaded Language	2,123	7.26%
Appeal to Fear	294	5.74%
Causal Oversimplification	209	4.84%
Appeal to Authority	144	4.69%
Name Calling	1,058	4.09%

Table 2. *Per-technique span F1 on SemEval-2020 evaluation set. Results from author's own experiments.*

Per-technique span F1 reflects both label quality and dataset imbalance. Gold span counts indicate the number of annotated instances per technique and highlight substantial class imbalance, which contributes to variability in performance. Techniques with stronger coverage achieve higher F1, while low-frequency techniques remain data-limited. Threshold sweeps show that most techniques favor low thresholds, reflecting recall-driven behavior and low-confidence predictions, with a few exhibiting clearer peaks.

5. Conclusion

This project constructed a Reddit-domain propaganda dataset of 173 high-consensus samples across 14 technique categories using active learning and LLM ensemble labeling. Dataset validity is confirmed by a DeBERTa classifier achieving Macro F1 of 0.817 on SemEval-2020.

References

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